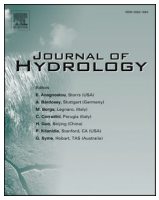
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Research papers



A flood Impact-Based forecasting system by fuzzy inference techniques

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ABSTRACT

The increasing frequency of severe floods worldwide highlights the need for effective communication and dissemination of information about potential floods and their possible consequences. Modern technology, such as social media and early warning systems, can play a crucial role in informing the public about impending floods and their potential impact. Impact-Based Forecasting (IBF), conveying meteorological and hydrological data, has emerged as a promising approach to inform vulnerable communities about the potential impact of flood disasters and to mitigate associated risks for resilience enhancement and losses minimization. This study proposes a flood impact-based forecasting system (FIBF) that uses flexible fuzzy inference techniques to construct a relation matrix of likelihoods and impacts and estimate the level of risk and impact for inducing storms. We construct a three-layer fuzzy inference model to determine the flood impact based on the combined effects of hazard, exposure, and vulnerability and then establish a FIBF warning product (up to 36 h ahead) to timely address flood information for inundation-prone regions of the Kemaman River basin in Malaysia. The FIBF can be displayed using Google Maps, providing a spatial visualization of the flooding situation to help decision-makers better understand the extent and severity of the flooding, make informed decisions on emergency response and resource allocation, and take proactive measures to reduce the negative impacts of flooding. Thus, the developed FIBF can be especially helpful in mitigating the socioeconomic costs of flood hazards.

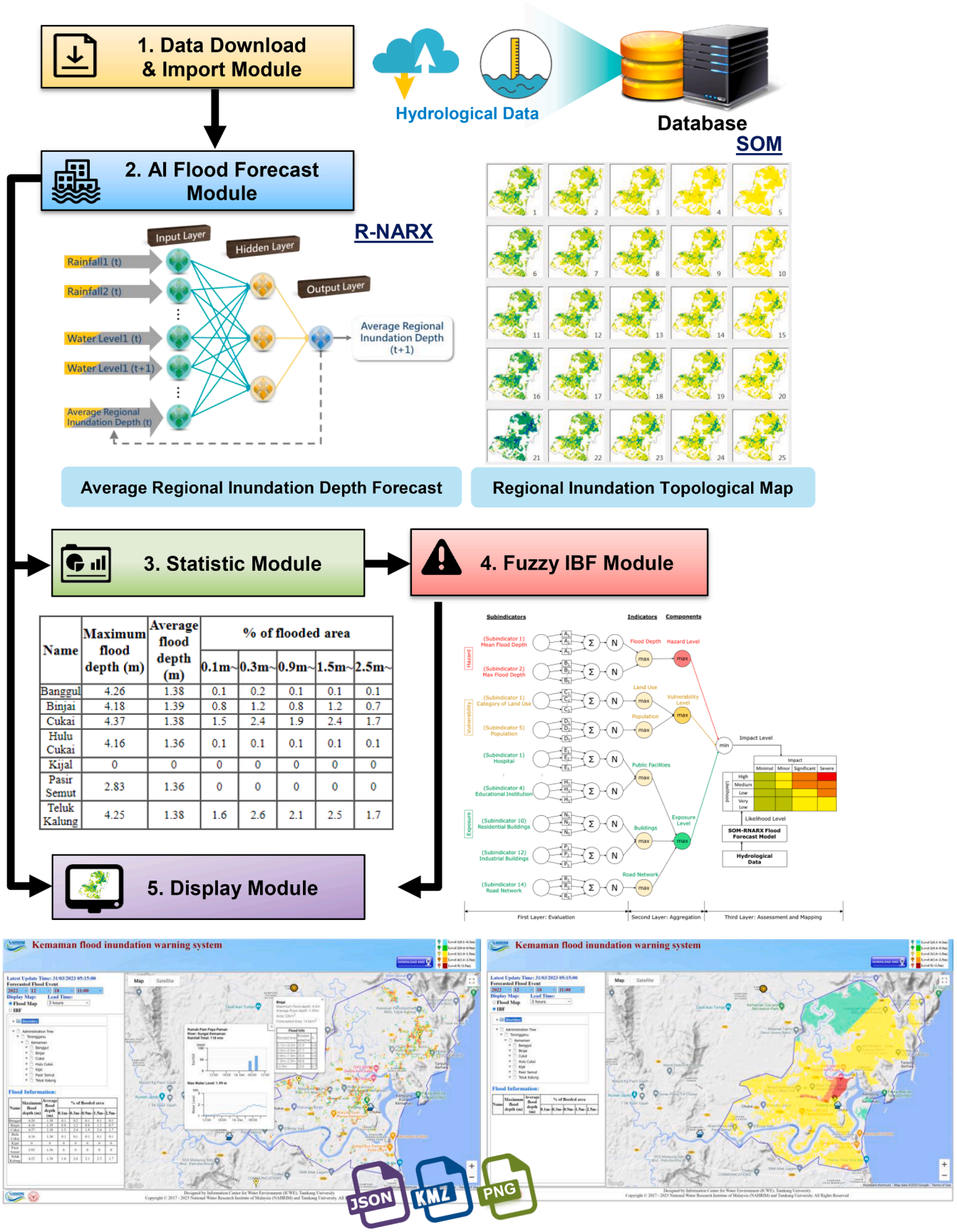
1. Introduction

Natural disasters are becoming more frequent and severe, causing devastating impacts on livelihoods, communities, and economies globally. Climate change and human-induced environmental changes are contributing to this trend by altering weather patterns, increasing the frequency and intensity of extreme weather events, and exacerbating existing vulnerabilities (e.g., Dong et al., 2021; Tabari et al., 2020; Wasko et al., 2021; Wu et al., 2022). Floods are the most common natural disasters, and the increasing trend of flood occurrence and severity has frequently been reported worldwide over the last decades (e.g., Guha-Sapir et al., 2011; Doocy et al., 2013; Li et al., 2021). Flood risk mitigation is a priority for many cities, particularly for those located in low-elevation coastal areas in Asia (e.g., Van Coppenolle & Temmerman, 2020; Wang et al., 2021; Chang et al., 2022; Scheiber et al., 2023). According to the World Disasters Report (McClean, 2010), Asia remains the most natural disaster-prone continent, with many regions highly vulnerable to the combined effects of subsidence and rising sea levels, which have already led to severe flooding. For instance, a series of unprecedented disasters, including the massive floods in Thailand in 2011, the 3/11 compounded disasters in Tohoku, Japan in 2011, and the 2013 Typhoon Haiyan in the Philippines, underscored the high impact of catastrophes on people and economic development (Aitsi-Selmi et al., 2016; Aldrich et al., 2015). These trends indicate the importance of community and government-based preparations and responses for reducing the damages raised from disasters. Decision-makers have recognized the vulnerability of the region to these shocks.

A crucial precautionary measure to cope with upcoming floods is to develop flood forecasts in each hotspot and warn the residents to reduce loss of life and property. Flood forecasts and warnings are essential tools to help people prepare and take necessary precautions to minimize the impacts of floods. Governments and non-governmental organizations play a vital role in providing essential resources and mechanisms to help people cope with flood disasters, such as emergency shelters, food, and medical supplies. Additionally, community-based initiatives can also help build resilience and preparedness among local populations. It is

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Name	Maximum flood depth (m)	Average flood depth (m)	% of flooded area				
			0.1m~	0.3m~	0.9m~	1.5m~	2.5m~
Banggul	4.26	1.38	0.1	0.2	0.1	0.1	0.1
Binjai	4.18	1.39	0.8	1.2	0.8	1.2	0.7
Cukai	4.37	1.38	1.5	2.4	1.9	2.4	1.7
Hulu Cukai	4.16	1.36	0.1	0.1	0.1	0.1	0.1
Kijal	0	0	0	0	0	0	0
Pasir Semut	2.83	1.36	0	0	0	0	0
Teluk Kalung	4.25	1.38	1.6	2.6	2.1	2.5	1.7

Fig. 1. Framework of the flood impact-based forecasting system. The system comprises five modules:(1) Data Download and Import module, (2) AI Flood Forecast module, (3) Statistic module calculates statistical values of regional forecasting inundation maps, (4) Fuzzy IBF module to assess the level of flood impact and to generate flood risk maps, and (5) Display module provided a user-friendly interface to visualize regional inundation maps and flood risk maps.

important to have effective mechanisms in place to prepare for and respond to flood disasters. This includes reliable flood forecasting systems that can provide advance warning to vulnerable communities, as well as emergency response plans and resources to help evacuate people or protect their homes and businesses. Higher levels of disaster preparation and deeper community awareness could mitigate damage arising from disasters in terms of the number of casualties and economic impacts (Davies et al., 2015; Dewan, 2015; Merz et al., 2020). An early flood warning system could enhance the preparedness of the communities and authorities for flash floods and mitigate the damage and risk for people in the vicinity of the river (e.g., Acosta-Coll et al., 2018; Cools et al., 2016; Perera et al., 2020).

Artificial intelligence (AI) techniques, such as machine learning and fuzzy inference, have proven to be efficient tools for learning nonlinear input–output systems and extracting significant features from complex data structures. In recent years, they have been increasingly used in hydrology and related fields to improve predictions and decision-making. Machine learning algorithms can learn patterns and relationships from large amounts of data, while fuzzy inference can handle imprecise and uncertain information. These techniques have been successfully applied in hydrological forecasting to enhance prediction accuracy and reduce uncertainty. The integration of AI techniques in hydrological forecasting can provide valuable insights for water resources management and disaster risk reduction (e.g., Nourani et al., 2014, 2021; Chang et al., 2019; Cheng et al., 2020; Kao et al., 2020; Lin et al., 2021; Zhou et al., 2019) and regional flood inundation forecasts (e.g., Chang et al., 2014; Chang et al., 2022; Kao et al., 2021; Ke et al., 2020; Nevo et al., 2022; Zhou et al., 2021). By leveraging these techniques, it is possible to make more accurate and reliable predictions, thereby helping to reduce the impact of natural disasters on both human populations and the environment. Effective real-time flood forecasting models can play a crucial role in mitigating flood disasters by rapidly disseminating inundation information for areas at risk. Furthermore, flood inundation maps provide policymakers with a clear depiction of the extent of the flood, effectively conveying the severity of different flood impacts and the effectiveness of potential countermeasures (Agnihotri et al., 2019; Chang et al., 2014; Sivanpillai et al., 2021; Tripathi et al., 2020; Zheng et al., 2018).

Recent studies have conducted hazard-focused warnings to communicate imminent extreme weather conditions and provide real-time urban flood information (Hemingway and Gunawan, 2018; Joe et al., 2021; Sufri et al., 2020). However, more than accurate flood forecasting is needed to take appropriate actions against flooding due to a lack of impact awareness. Thus, it is desired and crucial to convey metro-hydrological datasets to predict the impact of impending disasters and effectively communicate with people about the flood risks and relevant potential consequences (Demuth et al., 2012; Hemingway and Robbins, 2020). A growing body of flood-hazard knowledge about risk interprets and uses information in decisions during severe storms. The High Impact Weather (HIW) Project launched by the World Meteorological Organization (WMO) in 2016 is an important initiative to increase resilience to high-impact weather worldwide (WMO, 2015). One of the key components of this project is impact-based forecasting (IBF), which integrates weather information with vulnerability and exposure profiles to identify the level of risks in an area and adopt adequate responses to mitigate losses. Unlike traditional weather forecasting, IBF focuses on predicting the expected impact of a flood, rather than merely describing the weather status. This approach allows organizations and individuals to anticipate and take action to mitigate the impacts brought by storms, and thus minimize the socioeconomic costs of disasters and increase community resilience. The IBF approach provides qualitative impacts expected for each weather type and impact level, and its suitability has been recently investigated (e.g., Bierens et al., 2020; Boult et al., 2022; Nkiaka et al., 2020; Rözer et al., 2021; Sai et al., 2018). However, it requires quantifying the processes that come together and cause the final impact, which can be challenging. Furthermore, insufficient data for determining the relationship between damage and weather conditions would be another barrier to estimating potential levels of impact from a hazard.

This study aims to develop a comprehensive flood impact-based forecasting (FIBF) system for the Kemaman River basin in Malaysia, which integrates hydro-meteorological forecasts and high-resolution regional flood inundation depths with vulnerability and exposure profiles to create a warning product. The FIBF approach will utilize AI-based hydro-meteorological forecasts, a fuzzy reference warning system, and user-friendly visual presentations to provide flood risk information to various stakeholders. By monitoring rainfall and flood-prone areas, the FIBF will assess the risks and impacts of flooding on the affected region, including disruptions to production, social structures, and the environment. The development of the FIBF system will involve the construction of a flood early warning system and the implementation of a flood impact-based forecasting system in the inundation-prone regions of the Kemaman River basin.

2. Methodology

The study developed a web-based FIBF system comprising five major modules, namely, (1) Data Download Module and Import Module, (2) AI Flood Forecast Module, (3) Statistic Module, (4) Fuzzy IBF Module, and (5) Display Module. The methodology of constructing the FIBF system is shown in Fig. 1. The Data Download and Import Modules download real-time rainfall data from DID and import relevant hydrological data into the AI-based flood inundation forecasting model. The AI models generate regional multi-step-ahead flood forecast maps, while the Fuzzy IBF module calculates the impact of flooding on villages in the study area and the corresponding likelihood. The regional flood forecast maps and FIBF maps can be displayed online on Google Maps, forming a warning product that provides flood information for various users. A detailed description of the study area and developed models is shown in the subsequent sections.

2.1. AI-based model for regional flood inundation forecasts

An AI-based model has been developed for multi-step-ahead forecasting of regional inundation maps during flood events. The SOM-R-NARX model, proposed by Chang et al. (2014), is a hybrid approach that effectively combines the clustering capabilities of the self-organizing map (SOM) with the time series forecasting power of the recurrent configuration of nonlinear autoregressive with exogenous inputs (R-NARX). The methodology involves three main steps:

1. Clustering Inundation Maps: The SOM is used to categorize regional inundation maps into a meaningful topology, visualizing the inter-relational features of inundation depths under different hydrological conditions during storm events. The optimal size of the SOM can be determined by the coverage ratio between the maximum and minimum weights of neurons (Chang et al., 2021).
2. Forecasting ARIDs: The R-NARX network predicts the average regional inundated depth (ARID) during the storm event, using model outputs as inputs to handle the lack of real-time observed data, thus ensuring the continuity of forecasting capabilities. The optimal R-NARX network (i.e. the optimal number of neurons) is determined by minimizing the RMSE and maximizing the R-squared during the training and validation phases.
3. Adapting Inundation Maps: This step involves integrating the results from the SOM and R-NARX networks by adjusting the weights of the selected neuron in the SOM based on the forecasted ARID from the R-NARX. This refinement provides an updated inundation map.

The constructed SOM-R-NARX model can be repeatedly implemented for multi-step-ahead forecasting. It is worth noting that real-time rainfall information is the only data available for modelling in

IBF Risk Matrix		Impact			
		0-1	1-2	2-3	3-4
		Minimal	Minor	Significant	Severe
Likelihood	High				
	Medium				
	Low				
	Very Low				

Fig. 2. The flood-risk matrix of impact-based forecasting.

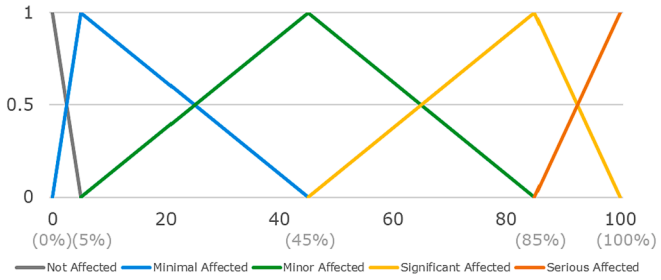


Fig. 3. The example of fuzzy sets of a sub-indicator.

real applications. Therefore, the proposed methodology provides an effective and promising way for forecasting regional flood inundation maps. This approach enhances flood preparedness and response strategies by providing visible regional flood inundation map forecasts. For more details on the constructed SOM-R-NARX model, refer to Chang et al. (2014) and Chang et al. (2022).

2.2. Development of impact-based forecast module

The World Meteorological Organization (WMO) guidelines on Multi-hazard Impact-based Forecast and Warning Services provide essential concepts and technical terms for impact-based forecasting (IBF) (WMO, 2015). These guidelines help to distinguish between a weather warning and an impact-based warning, which includes the vulnerability of people, livelihoods, and property in consideration of the hydrometeorological hazard (Harrowsmith et al., 2020). IBF is a useful approach to help people prepare for and mitigate the potential impacts of natural disasters such as floods. To develop an effective IBF system, risk and impact assessments are essential. This involves analyzing the potential impacts of weather events on vulnerable populations and infrastructure, as well as assessing the potential economic and social consequences of these impacts. By integrating this information into the IBF system, decision-makers can better understand the risks and potential impacts of weather events and take appropriate actions to mitigate these impacts and increase community resilience.

An IBF approach to forecasting a flood event might consider factors such as the likelihood and severity of flooding, the number and type of people and assets exposed to the flooding, and the vulnerability of those people and assets to flood damage. By integrating these factors into a risk assessment, an IBF model can provide accurate and relevant information to emergency responders, local government officials, and the general public. This information helps them prepare for and mitigate the potential impacts of the flood. Risk is defined as the probability and the hazard to human beings, livelihoods, and properties because of their exposure and vulnerability. Thus, risk assessments require information on hazard, exposure, and vulnerability. A hazard threatens life, property, and environment (Birkmann, 2006; UNDHA, 1992). Exposure refers to who and what may be affected in an area prone to hazards. Vulnerability represents susceptibility which means the degree of suffering to which people, livelihoods, and property are exposed when being affected by a hazard (Cardona, 1990).

The risk matrix shows the level of impact against the likelihood of impacts, which is an effective visual aid to interpret the combination of hazard forecast, vulnerability, and exposure data. It combines probabilistic hazard information with vulnerability and exposure data to produce impact forecasts. The level of impact combines forecasted hazard magnitude with exposure and vulnerability to calculate the potential severity of the impact. It can conceptually be presented in the following equation.

$$Level\ of\ impact = Hazard \times Vulnerability \times Exposure$$

The likelihood of impact considers the uncertainties in both the hydrometeorological forecast and its vulnerability and exposure to the potential severity of the impact. Fig. 2 illustrates the flood impact-based forecast that combines the impact with likelihood to create a risk matrix to express risk through a four-color scheme. The likelihood represents the levels of probability that a specific consequence, flooding, will occur—the longer the forecast time-step, the higher the forecast uncertainty. Thus, the likelihoods of forecast time steps for $T+3$, $T+12$, $T+24$, and $T+36$ are assigned as high, mid, low, and very low, respectively.

2.3. Fuzzy inference system (FIS)

A FIS is a powerful tool that integrates the fuzzy set theory, fuzzy IF-THEN rules, and fuzzy inference theory to interpret the input and allocate corresponding inferences to the output based on given rules. The IF-THEN rules are the core of a FIS and can be extracted from historical data or generated from human knowledge or other crisp rules through learning algorithms. These rules are typically expressed in the form of "IF (condition), THEN (consequence)," where the condition and consequence are represented using fuzzy sets. The FIS then uses these rules to interpret the input variables, which may be imprecise or uncertain, and allocate a corresponding output value based on the given rules. This allows the FIS to make accurate predictions and decisions even when the input data is incomplete or ambiguous. Fuzzy inference systems have a wide range of applications, including weather forecasting and flood prediction (e.g., Chang et al., 2005; Lu et al., 2014). By using historical data and expert knowledge to generate fuzzy rules, FIS can help decision-makers to make informed decisions and take appropriate actions to mitigate the impacts of natural disasters such as floods. Based on fuzzy IF-THEN rules, the inference engine maps fuzzy input sets into the output of fuzzy sets through fuzzy operations. Triangular norms and conforms (abbreviation T-norm and S-norm) are binary operations commonly used in fuzzy set theory to generalize the logical conjunction and disjunction. The T-norm operators are used for the intersection of fuzzy sets, while the S-norm operators are used for the union of fuzzy sets. This study uses the minimum and maximum operators to aggregate sub-indicators and indicators into a corresponding component.

There are five membership functions for every sub-indicator. Users can define the position of the membership function. Fig. 3 shows an example of fuzzy sets of a sub-indicator, where the maximum number of the sub-indicator is 100. The intervals between five peaks are 5%, 40%, 40%, and 15%, and the peaks of five triangular membership functions are 0, 5, 45, 85, and 100. The horizontal axis represents the values of sub-indicators, and the vertical axis represents the membership value of the fuzzy set.

2.4. Evaluation of model performance

The performances of the proposed approach are evaluated by the coefficient of determination (R^2) and the root mean square error (RMSE), as shown below.

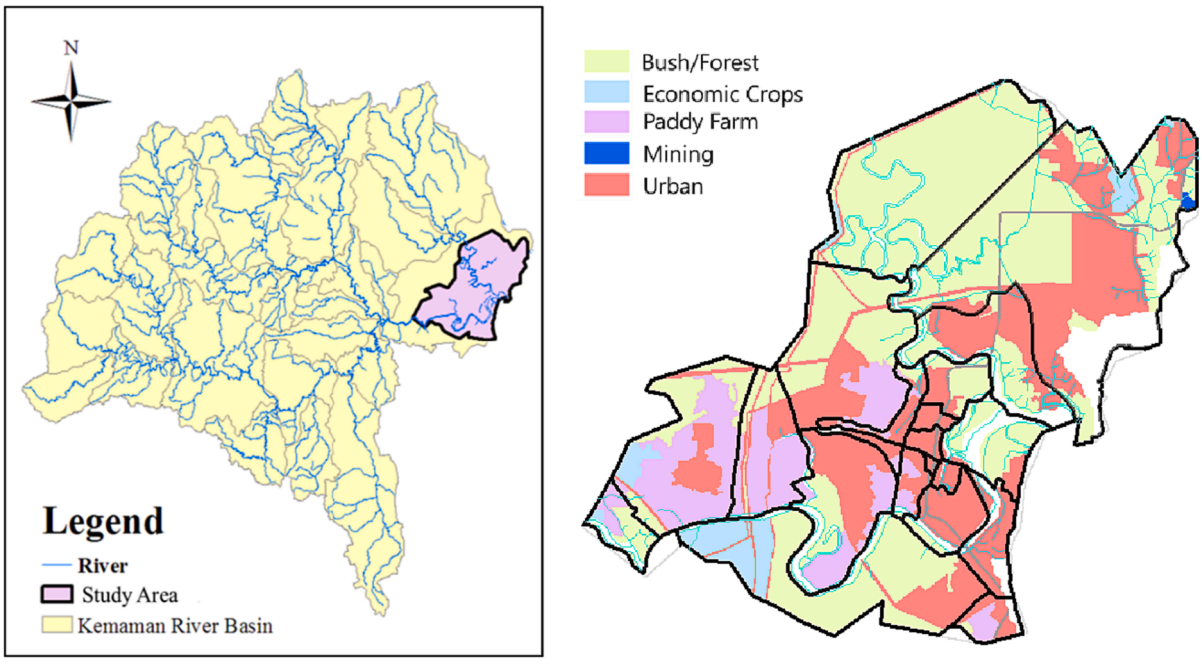


Fig. 4. Study area and five categories of land use.

Table 1
List of the primary data.

Data type	Method and source	Scale
Digital Elevation Model (DEM)		75 m × 75 m
River tributaries	JUPEM[2]	
Soil	MOA[3], 2010	
Land use	JPDB Terengganu[4]	
Building (Noncategorized)		
Discharge data	DID[1]	15 min
Rainfall data	DID	15 min

[1] DID: Department of Irrigation and Drainage, Ministry of Natural Resources, Environment and Climate Change, Malaysia.
[2] JUPEM: Department of Survey and Mapping, Malaysia.
[3] MOA: Ministry of Agriculture, Malaysia.
[4] JPDB Terengganu: Terengganu Town and Rural Planning Department, Malaysia.

Table 2
List of the data for assessment of risk matrix of IBF.

Data type	Category
Land use	Bush & Forest; Paddy farm; Economic crops; Mining; Village, Suburban, Urban
Buildings	Villages, Residential, Commercial, Industrial buildings
Population	Village population
Public facilities	Hospital, Clinic, Police station, Educational institution, Worship places, Government offices, Community halls, Evacuation centre
Transportation infrastructure	Jetty
Road network	Road network

$$R^2 = \left[\frac{\sum_{i=1}^{N}(D_i - \bar{D})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{N}(D_i - \bar{D})^2 \sum_{i=1}^{N}(Y_i - \bar{Y})^2}} \right]^2 \quad (8)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N}(D_i - Y_i)^2}{N}} \quad (9)$$

Table 3
Dataset allocation for modeling.

Dataset	Real Events	Designed Events
Training	Event 3	100YearReturnPeriod_12HourRainfall
	Event 5	50YearReturnPeriod_24HourRainfall
	Event 6	20YearReturnPeriod_48HourRainfall
	Event 7	50YearReturnPeriod_48HourRainfall
		20YearReturnPeriod_72HourRainfall
		100YearReturnPeriod_72HourRainfall
Validation	Event 2	20YearReturnPeriod_12HourRainfall
	Event 4	20YearReturnPeriod_24HourRainfall
		100YearReturnPeriod_48HourRainfall
Testing	Event 8	50YearReturnPeriod_12HourRainfall
		100YearReturnPeriod_24HourRainfall
		50YearReturnPeriod_72HourRainfall

where Y_i and D_i are the forecasted value and the observed (simulated) value of the ith data, respectively. $\bar{Y}$ and $\bar{D}$ are the mean of forecasted and the observed values, respectively. The RMSE ranges from 0 to infinity, with lower values indicating more accurate forecasts. The R^2 ranges from 0 to 1, with values closer to 1 indicating better predictability of a model and values closer to 0 indicating poor agreement between model predictions and observations.

3. Study area and materials

Floods in Malaysia make great social-economic impacts and are considered threatening. Being located on the east coast of West Malaysia, the Kemaman River basin experiences frequent and severe floods. The study area is located downstream of the Kemaman River basin and encompasses an area of approximately 100 km^2, including seven mukims and twenty-three submukims. Land use within this area falls into five categories as depicted in Fig. 4: bush/forest; economic crops; paddy farm, mining; and urban. Details of the study area can be found in Chang et al. (2018). The primary data collected for constructing the AI-based models and the IBF module is shown in Table 1 and Table 2, respectively.

Table 4
Performance of 3- to 36-h-ahead ARID forecasting by using the R-NARX networks.

Forecasting Time-step	RMSE (m) Training	Validation	Testing	R^2 Training	Validation	Testing
3-h-ahead	0.28	0.30	0.34	0.92	0.90	0.90
12-h-ahead	0.35	0.37	0.36	0.88	0.85	0.88
24-h-ahead	0.50	0.49	0.51	0.75	0.70	0.76
36-h-ahead	0.83	0.86	0.89	0.42	0.36	0.38

4. Results and discussion

4.1. R-NARX forecasting results

Table 3 presents the datasets used for modelling the R-NARX. Table 4 shows the summarized results for 3- to 36-hour-ahead forecasting of the ARID regarding RMSE and R^2 with linear modification. R^2 values of 3-hour-ahead forecasting exceed 0.90 in all phases, which indicates the robustness and capacity of the SOM-R-NARX model to forecast ARIDs reliably. However, as the forecast time horizon increased, model accuracy reduced slightly. For 12-hour-ahead forecasting, R^2 values reach 0.88, 0.85, and 0.88 in the three phases, respectively, while RMSE values remain low, ranging from 0.35 to 0.37 m, which confirm the model's predictive efficiency. In contrast, 24-hour-ahead forecasting exhibits lower R^2 values, ranging between 0.70 and 0.76, and higher RMSE values, slightly increasing from 0.49 to 0.51 m. 36-hour-ahead forecasted results are the least accurate due to limited rainfall information, achieving the lowest R^2 values of 0.42, 0.36, and 0.38, and the largest RMSE values of 0.83, 0.86, and 0.89 m in the three phases, respectively, compared to those of 3-, 12-, and 24-hour-ahead forecasting. The results imply that the R-NARX flood forecasting model has the potential to enable early and timely defense in flood-prone areas. However, 36-hour-ahead forecasting yields much lower R^2 values, indicating limitations due to the restricted availability of rainfall information. Future work could address this limitation by incorporating more dynamic weather forecasting models for long-term predictions.

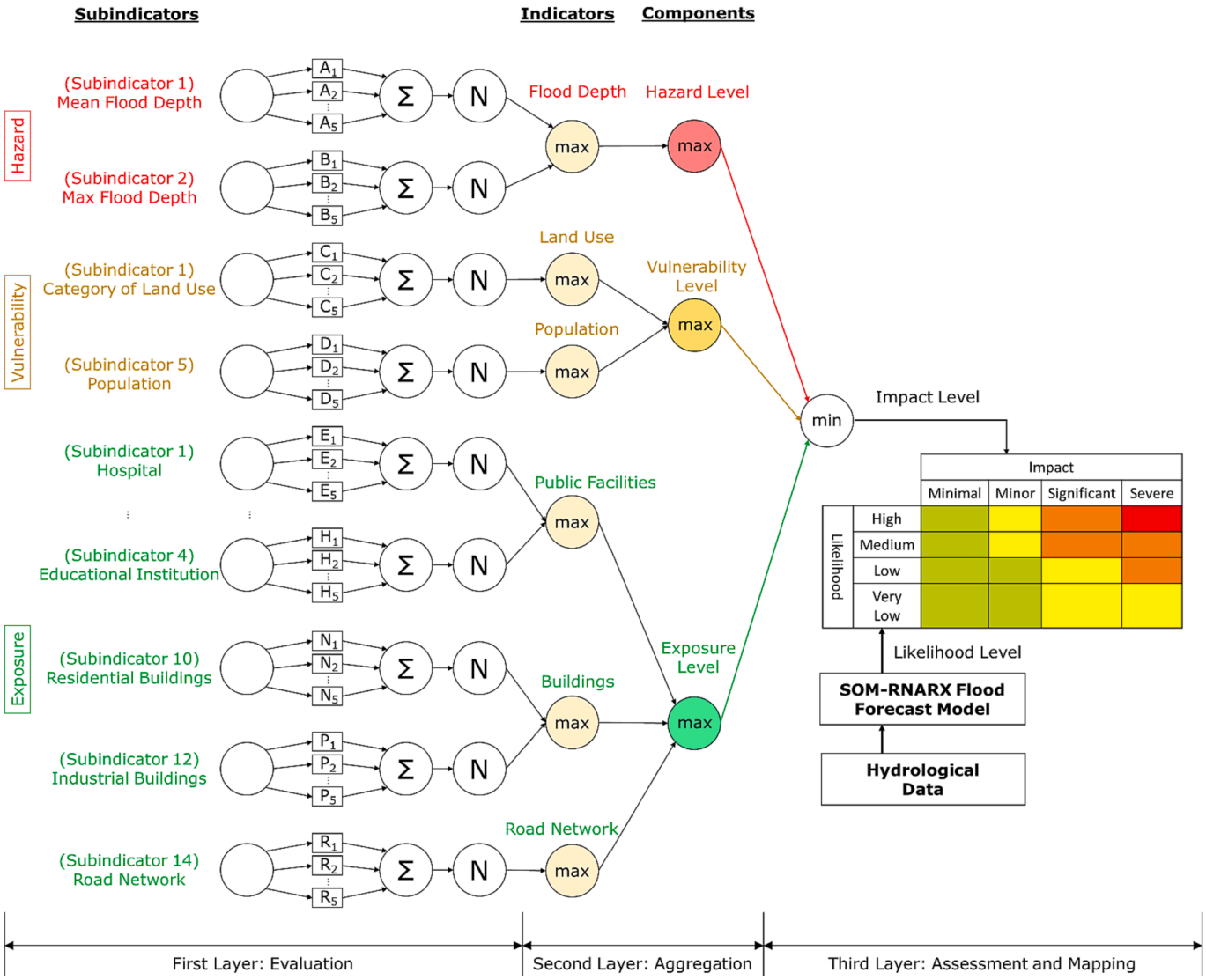
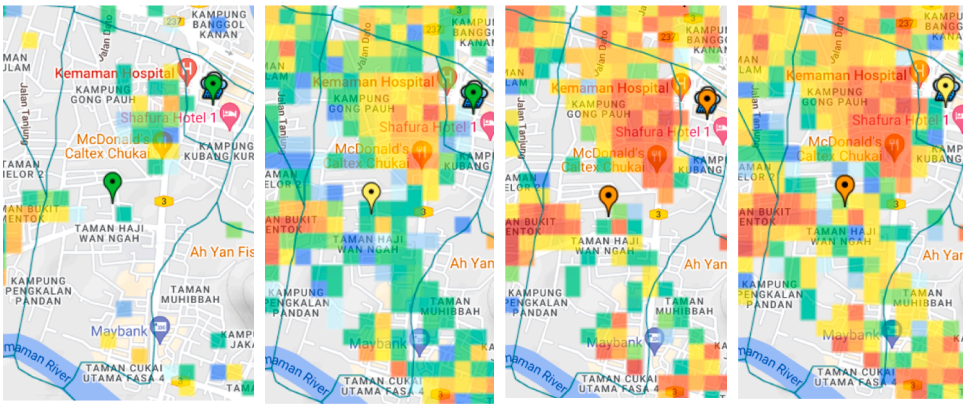


Fig. 5. Architecture of FIS for constructing FIBF.

Table 5
Indicators and sub-indicators of three components for the impact evaluation.

Component	Indicator	Sub-Indicator
Hazard	Flood depth	Maximum flood depth Mean flood depth
Vulnerability	Land use	Bush & Forest; Paddy farm; Economic crops; Mining; Village, Suburban, Urban
	Population	Population (from village shapefile)
Exposure	Public facilities	Hospital, Clinic, Police station, Fire station, Educational institution, Worship places, Government offices, Community halls, Evacuation centre
	Utilities	Electricity, Telecommunication
	Buildings	Villages, Residential buildings, Commercial buildings, Industrial buildings
	Transportation infrastructure	Bus terminal, Bridges, Jetty
	Road network	Road network



(a)3-hour-ahead (b)12-hour-ahead (c)24-hour-ahead (d)36-hour-ahead

Fig. 6. Forecast inundated maps of Kubang Kurus.

Table 6
Mean and maximum flood depths of different lead time forecast of Kubang Kurus.

Lead time	Mean flood depth (m)	Maximum flood depth (m)
3-hour-ahead	0.72	2.18
12-hour-ahead	0.95	3.76
24-hour-ahead	1.93	5.34
36-hour-ahead	1.58	4.94

4.2. Flood impact analysis

The flood impact analysis integrates hazard, exposure, and vulnerability assessments to provide a comprehensive approach for assessing flood risk. As presented in Fig. 5, this impact assessment leverages a three-layer fuzzy inference model to evaluate these three components using various indicators and sub-indicators presented in Table 5. The proposed three-layer fuzzy inference model provides a nuanced mechanism to derive flood impact, combining the influences of different indicators and sub-indicators.

The first layer, known as the Evaluation Layer, initiates the process. Here, each sub-indicator of the components is individually assessed, involving defining fuzzy sets, applying IF-THEN rules, and implementing defuzzification methods. The second layer, termed the Aggregation Layer, aggregates the outputs from the first layer, effectively consolidating the evaluated sub-indicators into their respective indicators and subsequently into their overarching components. The aim of the second layer is to derive the levels of hazard, exposure, and vulnerability. To

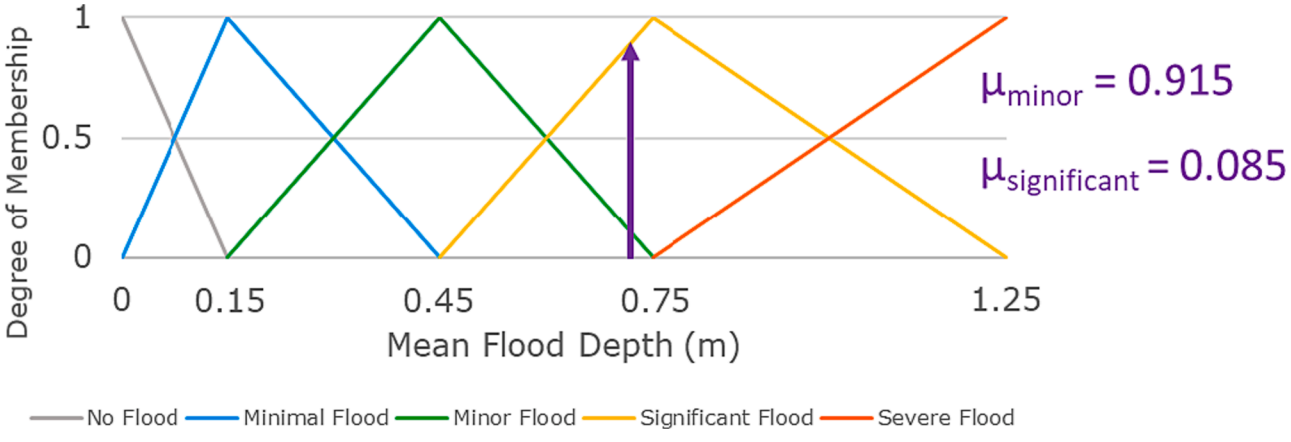


Fig. 7. The fuzzy set of the mean flood depth.

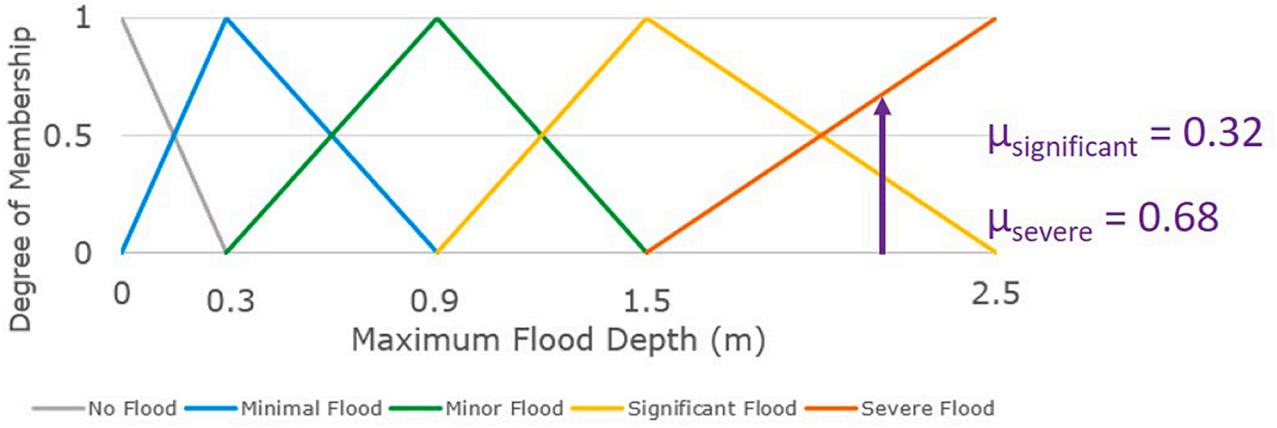


Fig. 8. The fuzzy set of the maximum flood depth.

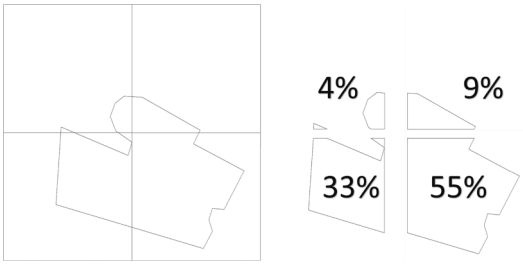


Fig. 9. Schematic diagram of a building spanning four grids.

Table 7
The number and the corresponding fuzzy outputs of affected indicators and sub-indicators of exposure in Kubang Kurus.

Indicator	Sub-indicator	3-h-ahead	12-h-ahead	24-h-ahead	36-h-ahead
Public facilities	Hospital	14.37 (2.37)	30(4)	30(4)	30(4)
	Clinic	0(0)	0(0)	0(0)	0(0)
	Police station	5.13 (3.45)	6.12(4)	6.12(4)	7.0(4)
	Educational institution	2.13 (1.12)	16.85 (2.83)	12.06 (2.27)	21.02 (3.8)
	Worship places	0(0)	2(4)	1(2.5)	2(4)
	Government offices	0.07 (0.2)	7.95(4)	7.64(4)	8.75(4)
	Community halls	0(0)	0(0)	0(0)	0(0)
	Evacuation centre	0(0)	1(4)	1(4)	1(4)
Building	Village	0(0)	1(2.5)	1(2.5)	2(4)
	Residential building	97.93 (1.64)	324.3 (4)	272 (3.0)	342.3 (4)
	Commercial building	20.01 (1.46)	85.7(4)	57 (2.55)	91.93 (4)
	Industrial building	0(0)	0(0)	0(0)	0(0)
Transportation	Jetty	0(0)	0(0)	0(0)	0(0)
Road network	Road network	4355 (1.58)	14,985 (3.84)	12,015 (2.83)	16199 (4)

*numbers in parentheses are the fuzzy outputs of sub-indicators.
*the unit of road network is meter.

ensure a comprehensive representation of the conditions affected by a specific indicator, an aggregation of fuzzy outputs is achieved using the maximum operator (S-norm). Finally, the third layer, termed the Assessment and Mapping Layer, operates the levels of hazard, exposure, and vulnerability obtained from the second layer using the minimum operator (T-norm) to yield the overall impact level. This impact level is

Table 8
The values of indicators of exposure in Kubang Kurus.

Indicator	3-h-ahead	12-h-ahead	24-h-ahead	36-h-ahead
Public facilities	3.45	4	4	4
Building	1.64	3	3	4
Transportation	0	0	0	0
Road network	1.58	3.84	2.83	4
Exposure	**3.45**	**4**	**4**	**4**

Table 9
The number and the corresponding fuzzy outputs of affected indicators and sub-indicators of vulnerability in Kubang Kurus.

Indicator	Sub-indicator	3-h-ahead	12-h-ahead	24-h-ahead	36-h-ahead
Land use population	Land use population	116 0	427 3335	342 3335	466 4184

Table 10
The values of indicators of exposure in Kubang Kurus.

Indicator	3-h-ahead	12-h-ahead	24-h-ahead	36-h-ahead
Land use	1.54	3.86	2.83	4
population	0	3.98	3.98	4
Vulnerability	**1.54**	**3.98**	**3.98**	**4**

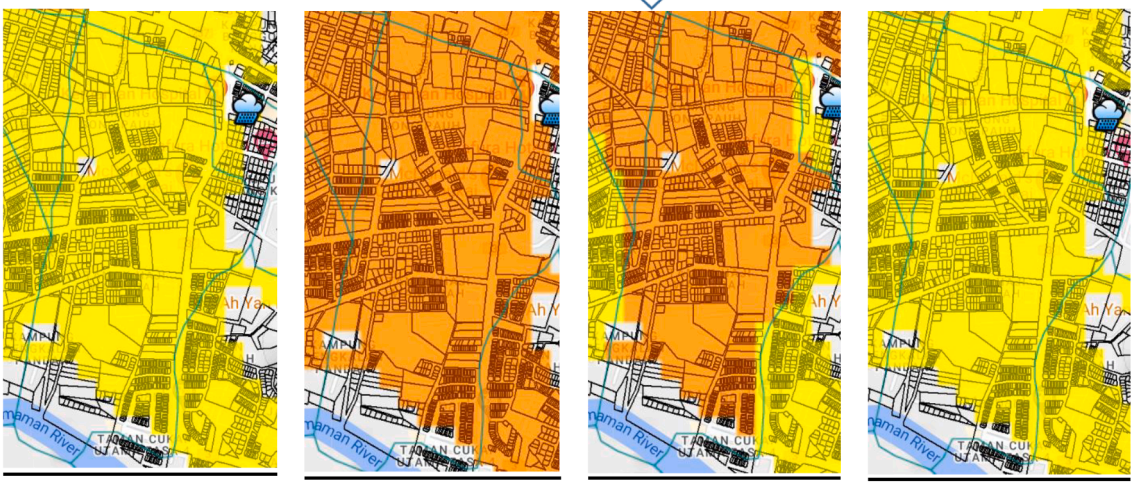
Table 11
The risk assessment in Kubang Kurus.

Component	3-h-ahead	12-h-ahead	24-h-ahead	36-h-ahead
Hazard	1.54	3.86	4	4
Exposure	1.58	3	4	4
Vulnerability	1.54	3.98	3.98	4
Impact	1.54(minor)	3. 86(severe)	3.98(severe)	4(severe)
Likelihood	high	medium	low	very low
Risk	A	B	C	D

then mapped against the likelihood in a risk matrix, producing a comprehensive flood risk map. The minimum operator plays an essential role in aligning the fuzzy sets within our FIS for constructing the FIBF model. It facilitates the aggregation of membership functions relating to hazard, exposure, and vulnerability.

Some parameters of FIS are pre-determined for constructing the Fuzzy based FIBF model. Firstly, it needs to identify the number of the fuzzy sets of each sub-indicator as well as the function-type of their membership functions. Secondly, it needs to select the type of fuzzy operators of sub-indicators, indicators, and components, respectively. Thirdly, it needs to set the outputs, known as the consequent part, of the fuzzy rules and determine the method of defuzzification.

IBF Risk Matrix			0-1	1-2	2-3	3-4
			Minimal	Minor	Significant	Severe
Likelihood	T+3	High		A		
	T+12	Medium				B
	T+24	Low				C
	T+36	Very Low				D



(a)3-hour-ahead (b)12-hour-ahead (c)24-hour-ahead (d)36-hour-ahead

Fig. 10. Flood risk maps of Kubang Kurus.

4.2.1. Hazard

To demonstrate the process of calculating the impact level by using the Fuzzy based IBF model and generating a flood risk map, Submukim Kubang Kurus is taken as an example. The process starts with executing the SOM-RNARX models to generate flood depths of all flooded grids of Kubang Kurus to calculate the mean and maximum flood depths. Fig. 6 shows 3-, 12-, 24- and 36-hour-ahead inundation maps. Table 6 shows 3-, 12-, 24- and 36-hour-ahead mean and maximum flood depths.

Fig. 7 and Fig. 8 show the fuzzy subsets of the mean and maximum flood depths, respectively. The method of defuzzification is a weighted average method. The defuzzifed values (fuzzy outputs) of the mean and maximum flood depths are called $Hazard_1$ and $Hazard_2$. The fuzzy if-then rules are defined as follows:

If the maximum flood depth is no flood, then the output is 0.
If the maximum flood depth is minimal, then the output is 1.
If the maximum flood depth is minor, then the output is 2.
If the maximum flood depth is significant, then the output is 3.
If the maximum flood depth is severe, then the output is 4.

At the mean flood depth of 0.72 m, the degrees of membership are 0.085 and 0.915 at significant and severe subsets, respectively, as shown in Fig. 7. Then, $Hazard_1$ is 2.91 as shown in Equation (1). Similarly, $Hazard_2$ is shown in Fig. 8 and Equation (2).

$$Hazard_1 = \frac{\mu_{minor} \times value\ of\ minor + \mu_{significant} \times value\ of\ significant}{\mu_{minor} + \mu_{significant}} = \frac{0.085 \times 2 + 0.915 \times 3}{0.085 + 0.915} = 2.91 \quad (1)$$

$$Hazard_2 == \frac{\mu_{significant} \times value\ of\ significant + \mu_{severe} \times value\ of\ severe}{\mu_{significant} + \mu_{severe}} = \frac{0.32 \times 3 + 0.68 \times 4}{0.32 + 0.68} = 3.68 \quad (2)$$

Through computing the Aggregation, the value of the hazard is max (2.91, 3.68) = 3.68.

4.2.2. Exposure

The process of calculating the exposure level is similar to that of the hazard level. Fig. 9 shows an example of calculating the coverage ratio of a flooded building in each grid when a building spans four grids. Table 7 shows the number of affected indicators and sub-indicators counted from the flooded grids in Kubang Kurus, and the sub-indicators fuzzy outputs calculated through fuzzy inference. Then, through computing the Aggregation, the values of indicators are shown in Table 8.

4.2.3. Vulnerability

The process of calculating the vulnerability level is similar to those of the hazard level and exposure level. Table 9 shows the number of affected indicators and sub-indicators counted from the flooded grids in Kubang Kurus, and the sub-indicators fuzzy outputs calculated through fuzzy inference. Then, through computing the Aggregation, the values of indicators are shown in Table 10.

4.2.4. Risk assessment

The values of three components (hazard, exposure, and

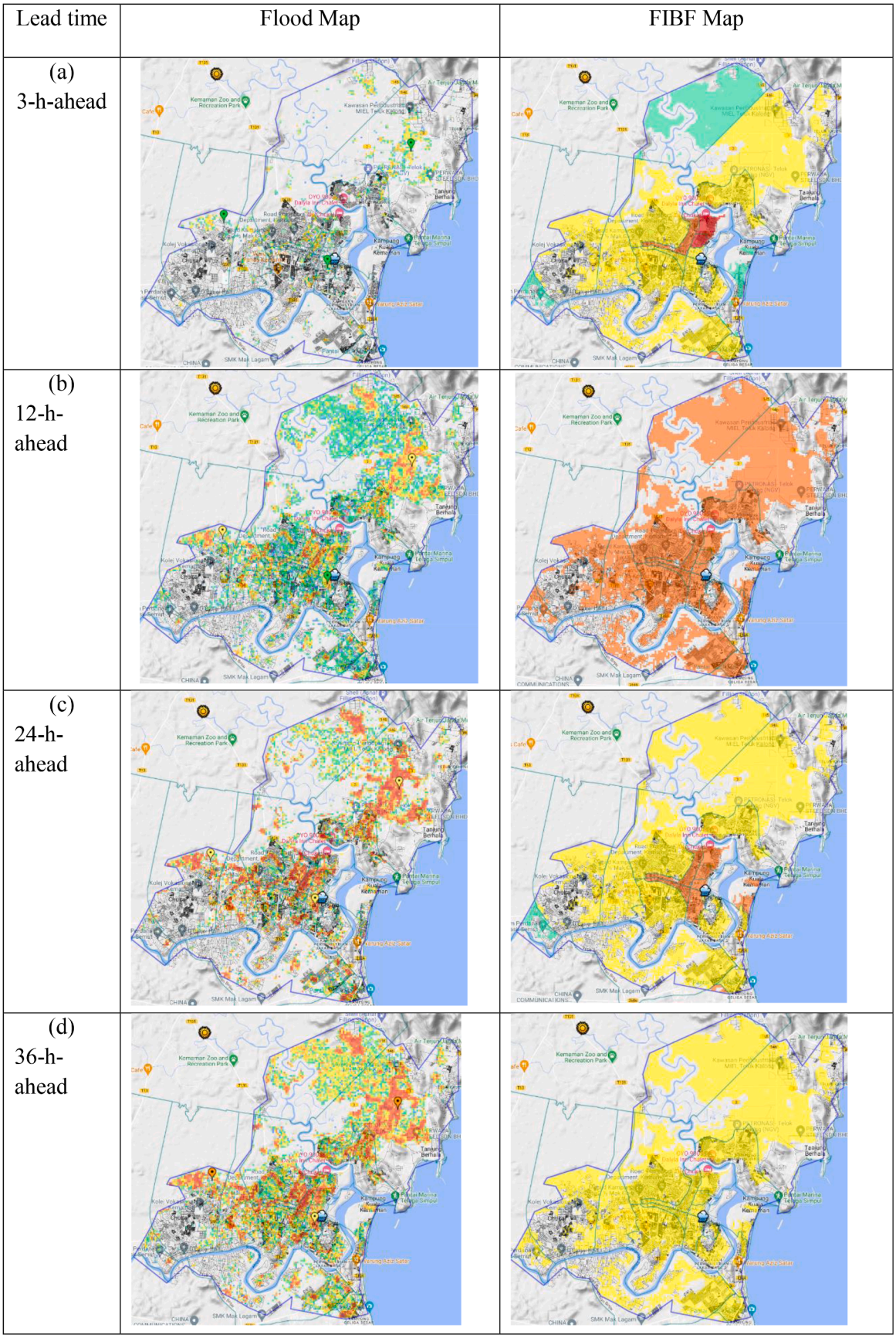


Fig. 11. Flood impact-based forecast maps at 02:00, 20/10/2022.

vulnerability) are aggregated to get the impact level by minimizing these three values, shown in Table 11. The results of risk assessment and the generation of a flood risk map, as shown in Fig. 10, reveal a practical application of our model. By mapping the likelihood and impact levels, our model could effectively communicate the flood risk to relevant authorities and communities, enabling appropriate preventative measures and response strategies.

4.3. Results of a flood impact-based forecasting system

Google Maps is a web mapping service that provides street maps. Our flood impact-based forecasting system was integrated with Google Maps, allowing for real-time flood inundation and impact-based forecast. This tool proves particularly valuable for decision-makers and emergency responders who need to quickly assess the potential impacts of a flood event. For an illustration of producing the flood impact-based forecast maps and using the risk matrix for the different levels of likelihood, 3-, 12-, 24-, and 36-hour-ahead flood forecast maps for the whole study area from AI models (SOM and R-NARX) at 02:00 on 2022 Oct 20th was selected as an example. We noted that the risk level decreased as the likelihood decreased given the same impact level, which suggests a lower risk for 24-hour-ahead forecasts rather than for 12-hour-ahead forecasts under the same impact level. Fig. 11 shows the flood impact-based forecast maps for 3-, 12-, 24- and 36-hour-ahead forecast results, respectively. The online platform displays the risk maps and provides a detailed forecast document for further technical insight.

In summary, the proposed methodology demonstrates how integrating fuzzy inference with spatial hazard information can enhance our understanding of flood risks. Our findings illustrate the potential of the proposed methodology in predicting flood risk, even though forecasting over extended periods remains a challenge. The use of a three-layer fuzzy inference model for impact assessment, coupled with the implementation of Google Maps for real-time visualization, offer a comprehensive and user-friendly platform for flood impact-based forecasting. This research paves the way for further refinement of these models, with the objective of improving long-term forecasting capabilities and incorporating real-time vulnerability and exposure data. We would like to note that the innovative model, in its current state, does have some limitations that require further efforts to enhance its applicability and transparency. These limitations are discussed as follows.

Subjectivity of fuzzy sets: The definitions of fuzzy sets and membership functions heavily rely on expert judgment, introducing a degree of subjectivity. While this allows for the integration of expert knowledge, it may lead to inconsistencies or bias in the system.

Data availability and quality: Fuzzy systems perform optimally when provided with high-quality and comprehensive data sets. In many real-world scenarios, obtaining such data, especially for parameters like vulnerability and exposure, can be challenging due to privacy concerns, limited resource, or incomplete record-keeping.

Temporal dynamics: The FIS, as commonly used, may not fully capture the temporal dynamics of flood exposure and vulnerability. These factors can evolve over time due to changes in demographics, land use, climate change, and other variables. As a result, the FIS may require periodic updates to reflect these changes accurately.

Despite these limitations, it is essential to emphasize that fuzzy inference systems remain a valuable tool for flood impact assessment and decision-making processes.

5. Conclusions

Natural disasters such as floods significantly impact people's livelihoods, often necessitating large-scale evacuations and disrupting manufacturing chains. Coastal cities in low-lying tropical regions are especially prone to flooding due to sea level rise and extreme weather events. As a result, flood risk mitigation becomes a paramount concern for these regions. This study presents a cutting-edge, web-based flood impact-based forecasting system (FIBF) that incorporates meteorological and regional flood inundation depth forecasts for providing flood information to various stakeholders. The FIBF system can help translate early warnings into early action and timely provide accurate and comprehensible climate information to minimize socioeconomic costs, save countless lives, and alleviate suffering caused by weather and climate hazards. The proposed methodology and main results are summarized as follows:

(1) Hydro-geologic datasets are collected, analyzed, and organized into a database using the Microsoft SQL Server. This database stores real-time hydrological data from Department of Irrigation and Drainage (DID, Malaysia), manages model-relevant data, stores forecasted data, and provides information for visual map-based displays.
(2) An AI-based flood inundation forecasting model is developed and tested, which provides accurate visible multi-step-ahead (up to 36 h ahead) regional flood inundation maps with low RMSE and high R^2 values.
(3) The FIBF risk map is derived from the AI-based forecasting system and successfully implemented in an online FIBF risk map.
(4) A web interface is designed to present online forecasts of flood inundation depths and FIBF in the study area. Built using the up-to-date web technologies and driven by database, this interface offers real-time updates and straightforward access to critical information. The display interface features various capabilities, including the depiction of the inundation area on Google Maps, provision of the maximum inundation depth, and the display of specific locations. This user-friendly tool equips decision-makers and the public with access to crucial flood forecasting and impact-based forecasting information.

The implementation of the proposed FIBF system in Malaysia marks a significant milestone in bolstering community resilience and preparedness against flooding. Incorporating impact-based forecasting into the national decision-making process can help lessen the adverse impacts of flooding and alleviate the socioeconomic costs associated with natural disasters. However, there is potential for system improvement. For example, integrating hydro-metrological forecasts with vulnerability and exposure data could yield more accurate flood forecasts and warnings. Additionally, ensuring clear, understandable communication of these forecasts and warnings with decision-makers, the public, and other stakeholders is vital. By continuing to develop and improve the FIBF system, we can further fortify our endeavor to translate early warning into early action, ultimately preserving lives and reducing the hardship caused by natural disasters.

Credit authorship contribution statement

Gary Wee: Conceptualization, Software, Methodology, Formal analysis, Visualization, Investigation, Writing – original draft. **Li-Chiu Chang:** Conceptualization, Methodology, Supervision, Writing – review & editing. **Fi-John Chang:** Conceptualization, Methodology, Writing – original draft, Writing – review & editing. **Mohd Zaki Mat Amin:** Conceptualization, Funding acquisition, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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